Database Management Systems - Normalization & Indexing

Course: Database Systems | Semester 4 | Contributor: Priya Sharma

Step-by-step decomposition examples with functional dependency preservation proofs.

Unit 1: Functional Dependencies & Closure

* Armstrongs Axioms: Reflexivity, Augmentation, Transitivity, and Union.
* Attribute closure computation and finding candidate keys algorithm.
* Canonical cover derivation for minimal functional dependency sets.

Unit 2: Normal Forms 1NF, 2NF, 3NF, BCNF

* 1NF: Atomic attribute values; 2NF: No partial dependency on candidate keys.
* 3NF: For X -> A, either X is superkey or A is prime attribute.
* BCNF: For every non-trivial dependency X -> A, X must be a superkey.
* Lossless join decomposition testing and dependency preservation criteria.

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